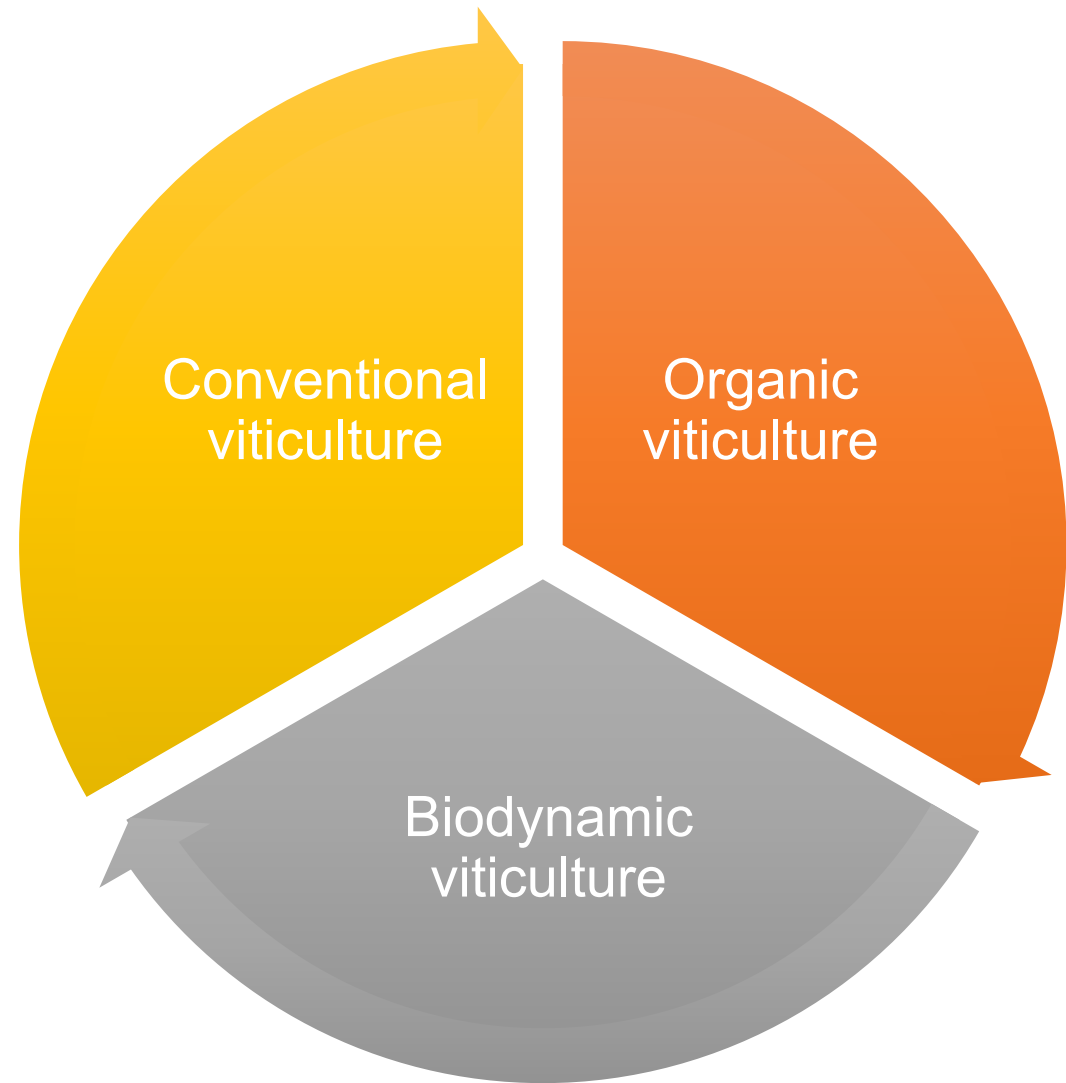


Fundamentals of modern viticulture

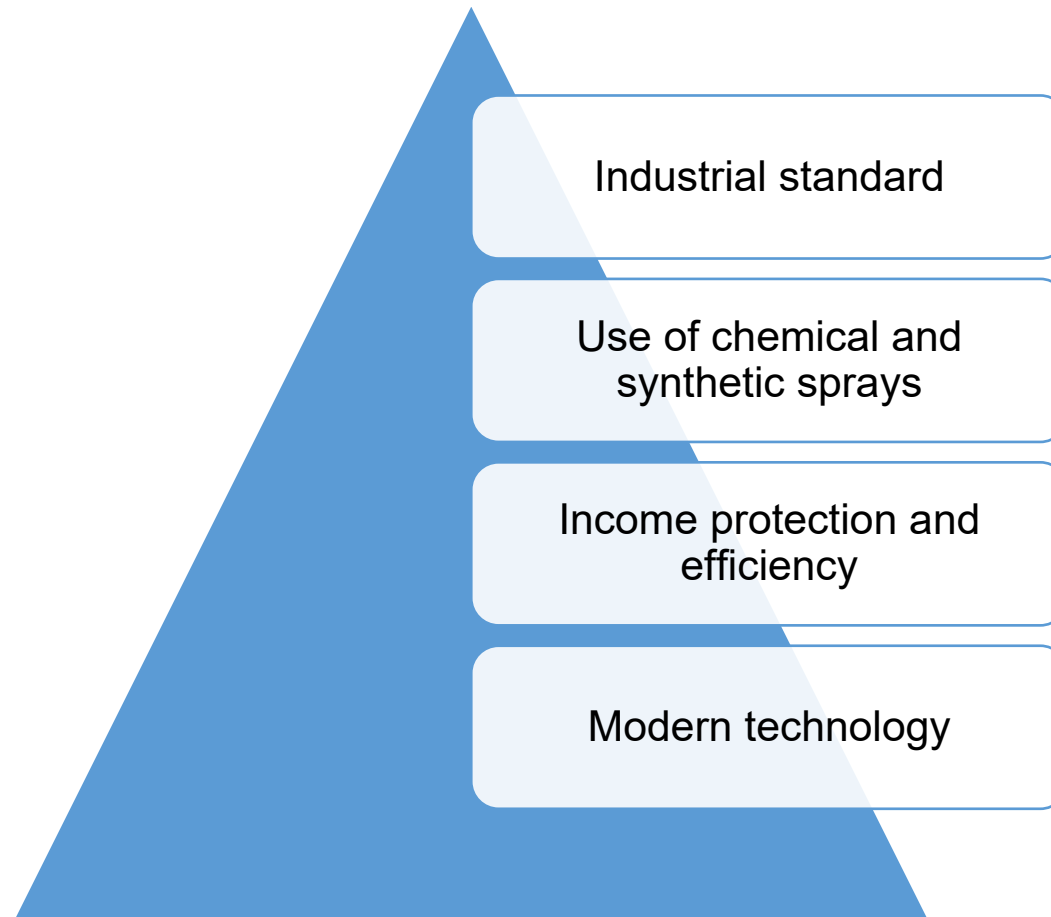
wine workshop

Types of management

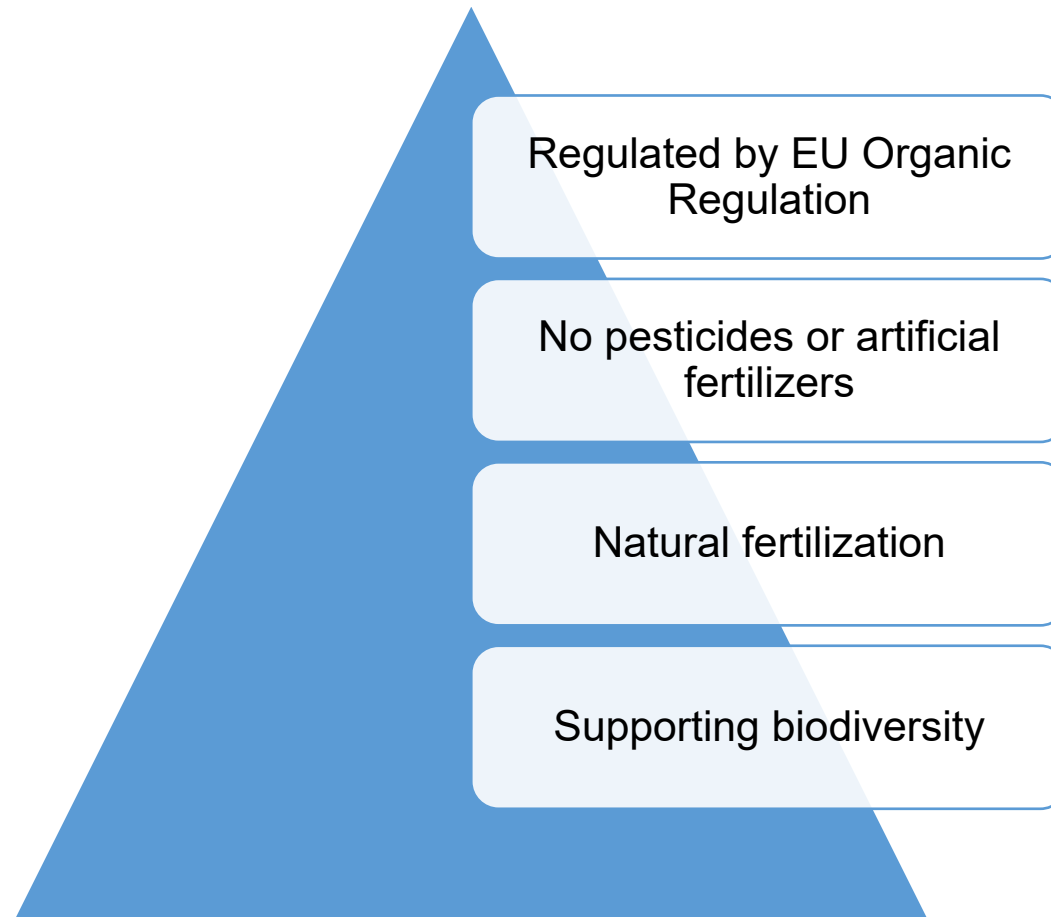




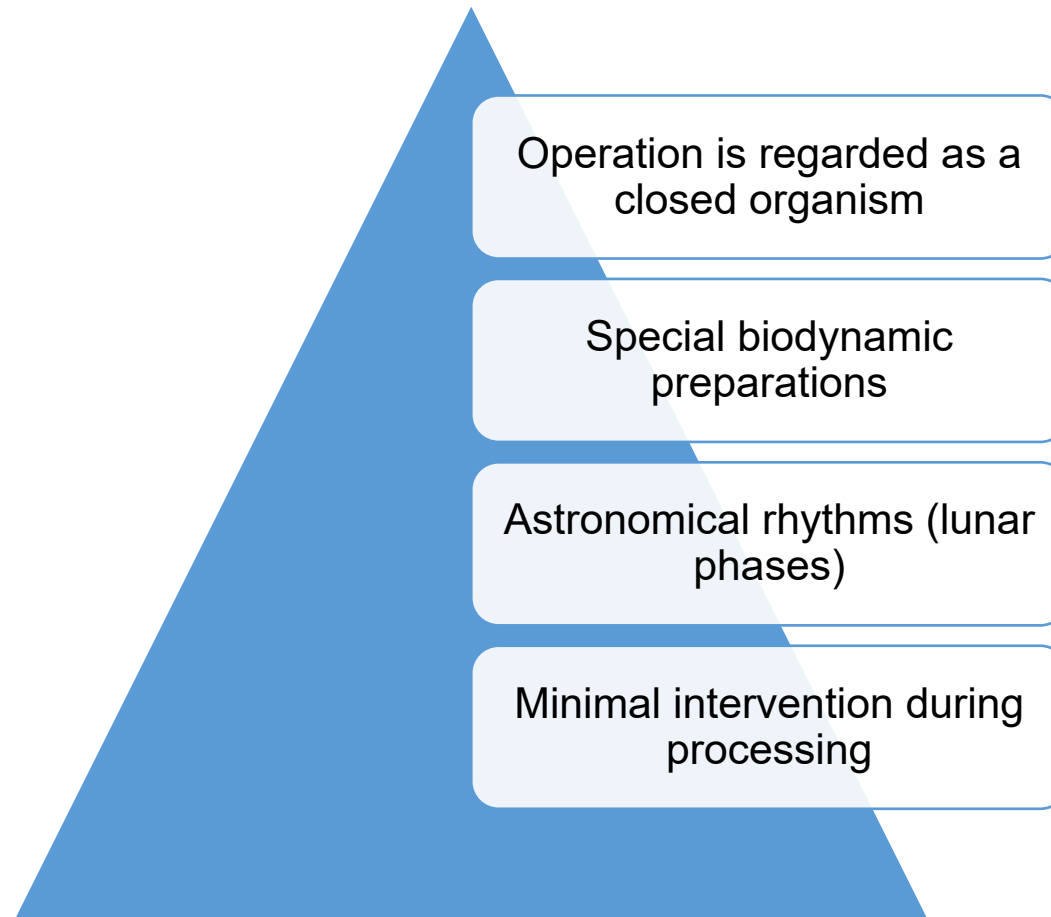
Conventional viticulture



Organic viticulture

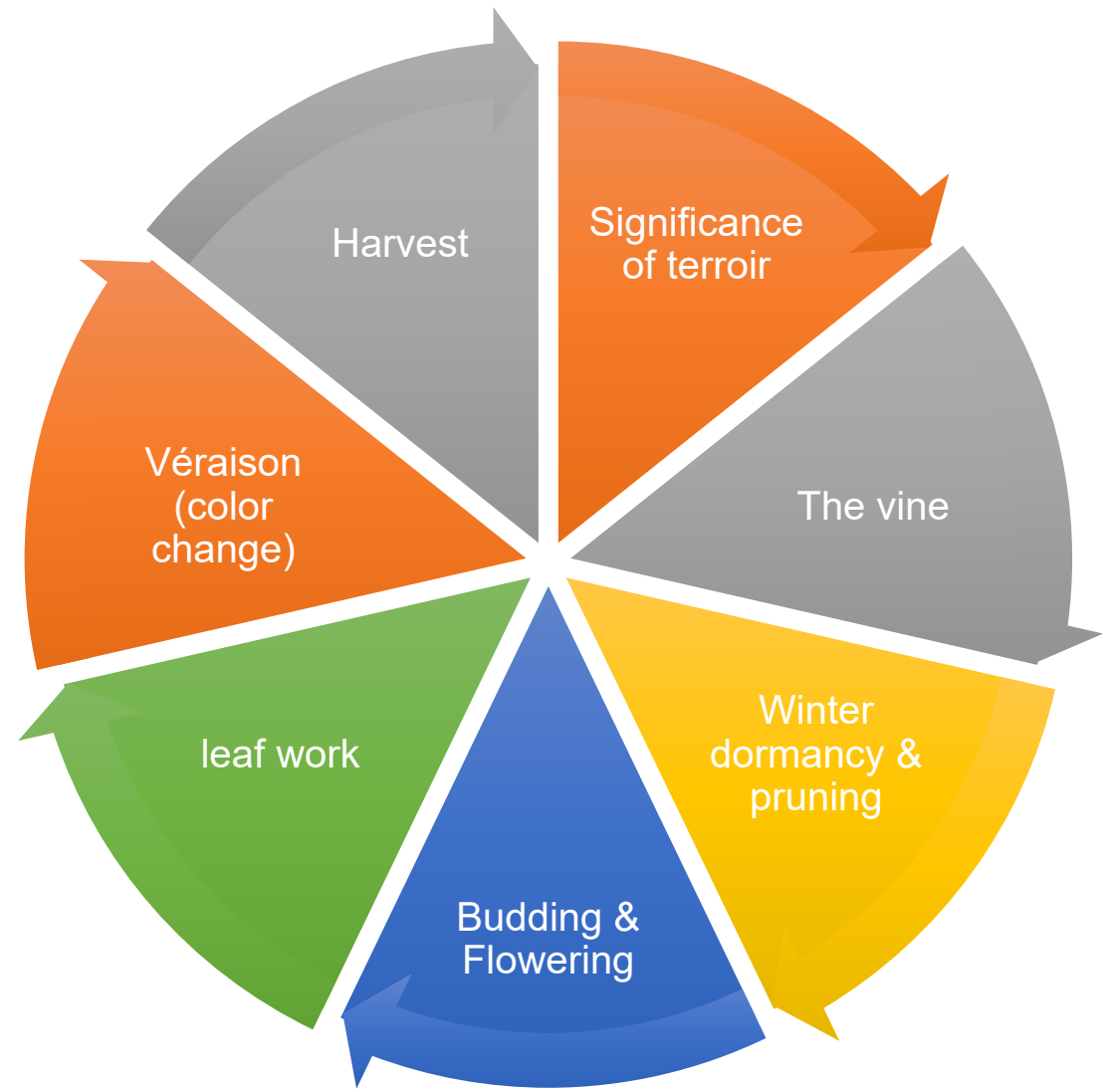


Biodynamic viticulture (Demeter)





The yearly cycle in the vineyard



The 4 pillars of terroir

climate

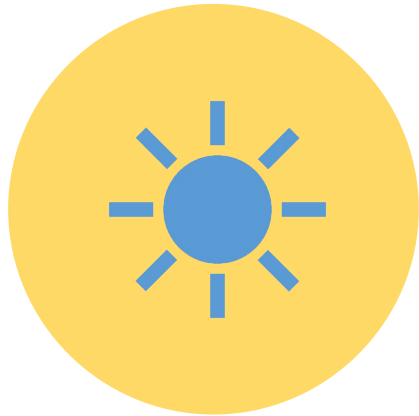
soil

location

tradition



The climate is the energy



SUNSHINE AND TEMPERATURE
INFLUENCE SUGAR CONTENT
AND AROMA DEVELOPMENT



PRECIPITATION INFLUENCES
GROWTH



COOL NIGHTS ENSURE ACIDITY

The floor is the foundation



EVERY FLOOR STORES WATER
AND HEAT DIFFERENTLY



VINES HATE WET FEET



DEEP ROOTS PROVIDE
RESISTANCE

Location is the key



EXPOSURE TO
SUNLIGHT



LEVEL OF AIR FLOW
& UV RADIATION



WATER REFLECTS
THE SUN

Tradition – the human factor



THE CHOICE OF GRAPE VARIETY



VINE CULTIVATION AND LEAF WORK



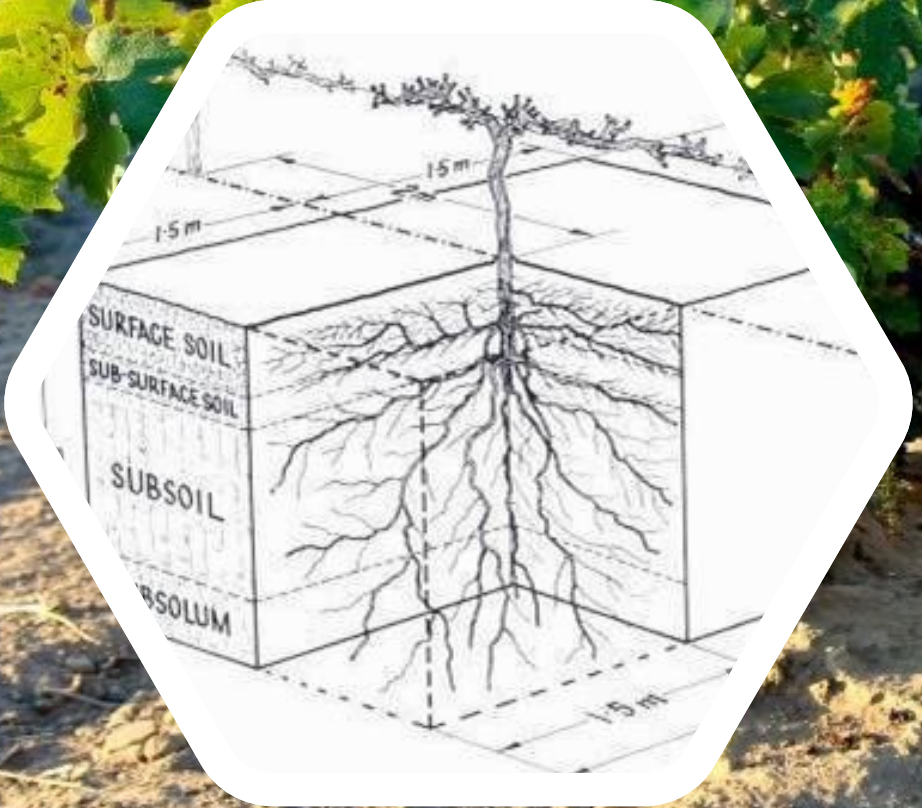
INCOME RESTRICTION

Structure of the vine

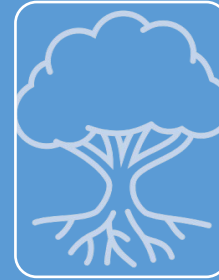
roots

plant

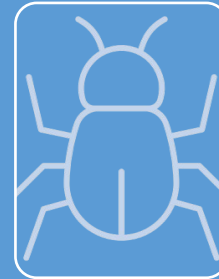
grape



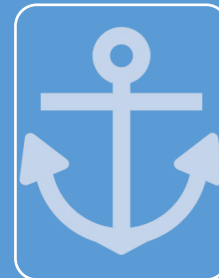
Roots the underground system



The vine is a deep-rooted plant, absorbing water and minerals.



The American wild grapevine is resistant to phylloxera.



The rootstock determines how the plant copes with drought and nutrients.

Plant the architecture



The trunk is the conduit system for nutrients.



Only the buds of the one-year-old wood grow fruit-bearing shoots in spring.



The leaves are the solar panels; they produce sugar through photosynthesis.



The inflorescences are the flower clusters of the vines; self-pollination produces the grapes.

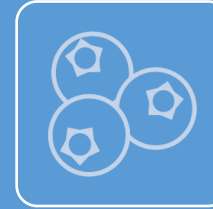


A grapevine is a vine that clings to the frame with its tendrils.

Grapes the fruit in detail



Rapeseed (stem structure) contains many tannins and is usually removed.



Berry skin, the source of colorants, aromas, and tannins



Fruit pulp consists mainly of water, sugar, and acid.



Seeds contain bitter compounds and oils.

forms of education

- Gobelet / bush management
- ground management
- Guyot system
- Kordon system
- single-pole method
- Pergola - system

Gobelet / bush management



Advantage:

- sun protection
- water management
- ground heat

ground management



Advantage:

- frost protection
- windshield
- ground heat
- moisture protection

Guyot – System



Advantage:

- sun exposure
- ventilation
- mechanization
- Easy care

Kordon – System



Advantage:

- wind stability
- Simple pruning
- mechanization
- Constant ventilation

single-pole method



Advantage:

- Farming on steep sites
- Optimal use of light
- ground heat
- Individual care

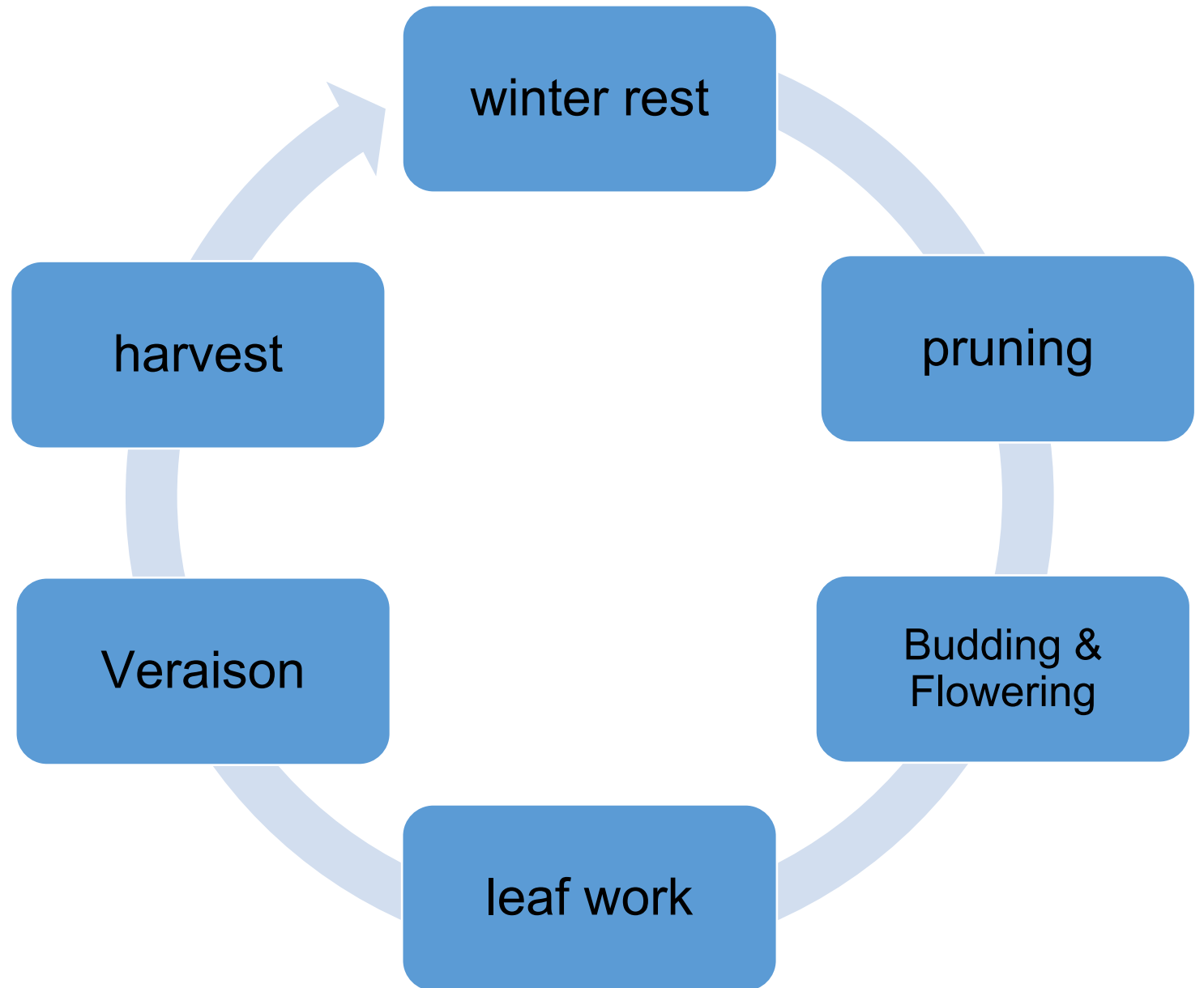
Pergola - System



Advantage:

- Protection from heat & sunburn
- Optimal use of space
- Ventilation in humid areas
- Comfort during harvesting

The yearly cycle





winter rest



Sap withdrawal: the plant withdraws nutrients from the shoots back into the old wood and roots.



Soil revitalization: during the dormant period, preparations are applied to stimulate root formation in the spring.



Frost: moderate cold reduces pests, but extreme frost can damage the wood



Repairs: Time to repair wire frames and posts in the vineyard

pruning



Growth control: Pruning determines the shape of the vine. The aim is to ensure good ventilation and sunlight exposure for the leaves.



Gentle pruning: Only cut into young wood, as this prevents large wounds and protects the sap channels.



Yield regulation: The fewer buds, the lower the quantity, but the higher the concentration and quality of the grapes.



Biodynamic timing: During the waning moon, as sap flow is at its lowest then.

Budding & Flowering



Bleeding: Before the first leaves sprout, sap oozes from the cut surfaces—a sign that the roots are actively drawing water again.



Wool stage & bud break: Buds begin to swell, break open, and the first green shoot tips emerge.



Self-pollination: The vine is hermaphroditic and does not require bees. Wind disperses pollen from the anthers to the stigma.



Drizzle: Rain or severe cold prevent the flowers from being fertilized, which significantly reduces the yield.

leaf work



Pruning: Removing superfluous shoots from the trunk and from inside the vine so that the energy flows directly into the fruit shoots.



Leaf removal: Targeted removal of leaves in the grape zone. Better ventilation or sun protection.



Topping (capping): Shortening the shoot tips in summer to stop length growth and redirect energy to the ripening of the berries.



Suckering management: Removal of side shoots (“suckers”) to prevent excessive foliage density.

veraison



Visual change: The grapes lose their green color (chlorophyll degradation).



Metabolism update: sugar increase, acid breakdown, aroma formation



Physical characteristics: skin becomes soft and elastic, volume increases due to water retention.



Significance for winegrowers: Start of ripeness monitoring, time for yield reduction

harvest



Determining the timing: measuring sugar ripeness (must weight), acid ripeness, physiological ripeness (aromas), ripe tannins



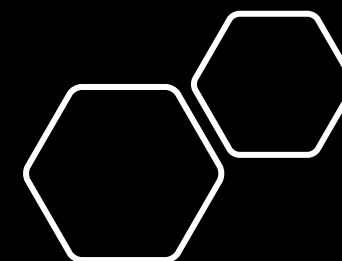
Weather: A dry period is ideal to prevent rot and dilution of the must by rain



Harvesting methods: Hand harvesting is gentle and quality-oriented, but costly and time-consuming. Machine harvesting is cost-efficient. Usually takes place in the early hours of the morning.



Logistics & cellar management: Process grapes quickly, ensuring they remain undamaged and are carefully selected. Prevent oxidation and wild fermentation.



Thanks